Skills Summary

Distinguishing Exponential from Linear Change

Use this reference while completing your worksheet or when studying.

One question decides everything: add or multiply?

- **Linear** — same amount *added* each step. Constant difference. $f(n) = a + d\,n$.
- **Exponential** — same factor *multiplied* each step. Constant ratio. $f(n) = a\,r^n$.
- Neither row constant? Then it is neither model. "Not linear" does not mean exponential.

Language clues: "per year," "each month it adds" → linear. "doubles," "percent of its current value," "times as many" → exponential.

1. Decide from a table: differences or ratios

Write *both* rows under the table. Subtract consecutive values for the difference row; divide consecutive values for the ratio row. Whichever row is constant names the model, and that constant is d or r.

Example: $f(n)$ is 3, 7, 11, 15 at $n = 0, 1, 2, 3$. Find $f(6)$.
Step 1. Test both rows before naming a model, since increasing values alone prove nothing.
Step 2. Differences $7-3$, $11-7$, $15-11$ are all 4 — constant — so f is linear with $a = 3$, $d = 4$, and $f(6) = 3 + 4 \cdot 6$.

$$f(6) = \mathbf{27}$$

Try it: $g(n)$ is 2, 6, 18, 54 at $n = 0, 1, 2, 3$. Decide the model, then find $g(5)$.

check: **486**

2. Write the rule from a description

Find the starting value a at step 0. Then read the verb: an amount added is d, a factor multiplied is r. "Doubles" is $r = 2$; "grows 20 percent" is $r = 1.2$; "loses 20 percent" is $r = 0.8$.

Example: A count starts at 5 and doubles each step. Find the value at step 4.
Step 1. "Doubles" is a repeated multiplication, so the model is exponential with $r = 2$ rather than a constant amount added.
Step 2. The rule is $f(n) = 5 \cdot 2^n$, so $f(4) = 5 \cdot 16$.

$$f(4) = \mathbf{80}$$

Try it: A count starts at 7 and triples each step. Find the value at step 3.

check: **189**

3. Compare a linear model with an exponential one

Early steps are misleading: the linear model is usually ahead at first. Compare by evaluating both rules at the same input and subtracting. An exponential model with $r > 1$ always overtakes any linear model eventually.

Example: $L(n) = 20 + 10n$ and $E(n) = 20 \cdot 1.5^n$. How far ahead is E at $n = 6$?
Step 1. Evaluate both rules at the same input and subtract, instead of comparing their step sizes.
Step 2. $E(6) = 20 \cdot 1.5^6 = 227.8125$ and $L(6) = 20 + 60 = 80$, so the gap is $227.8125 - 80$.

$$E(6) - L(6) = \mathbf{147.81}$$

Try it: With the same two rules, how far ahead is E at $n = 4$?

check: **41.25**

4. Classify a real situation, then compute

A percent *of the current amount* is exponential; a fixed quantity per period is linear. Losing 20 percent a year multiplies by 0.8 each year; losing 20 litres an hour subtracts 20 each hour. Classify before you compute — the arithmetic is easy, the choice is where marks are lost.

Example: A 600-dollar phone loses 20 percent of its value each year. What is it worth after 3 years?
Step 1. The loss is a percent of the current value, so each year multiplies by 0.8 — exponential decay, not a fixed subtraction.
Step 2. $V(n) = 600 \cdot 0.8^n$, so $V(3) = 600 \cdot 0.512$.

$$V(3) = \mathbf{307.2}$$

Try it: A 600-litre tank drains 20 litres every hour. How much is left after 3 hours?

check: **540**

(!) Watch out: $a \cdot r \cdot n$ is not $a \cdot r^n$ — multiplying by the number of steps is the linear move. And a percent taken from the *original* amount every period is linear; a percent of the *current* amount is exponential.